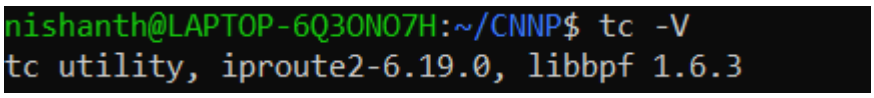
2520090141 CNNP SKILL 3

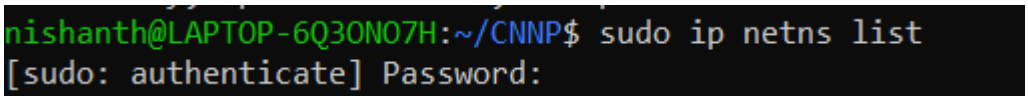
iPerf3:



Check Traffic Control:



Check your existing network namespaces:



Check Wireshark:

```
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ wireshark --version
Wireshark 4.6.4.

Copyright 1998-2026 Gerald Combs <gerald@wireshark.org> and contributors.
Licensed under the terms of the GNU General Public License (version 2 or later).
This is free software; see the file named COPYING in the distribution. There is
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compile-time info:
 Bit width: 64-bit
  Compiler: GCC 15.2.0
      GLib: 2.87.3
 With:
  +brotli                      +nghttp2 1.68.0
  +Gcrypt 1.12.0               +nghttp3 1.12.0
  +GnuTLS 3.8.12 and PKCS#11   +PCRE2 10.46 2025-08-27
  +Kerberos (MIT)              +POSIX capabilities (Linux)
  +libnl 3                     +Qt 6.10.2
  +libpcap                     +QtDBus
  +libsmi 0.4.8                +QtMultimedia
  +libxml2 2.15.1              +Snappy 1.2.2
  +Lua 5.4.8                   +xxhash 0.8.3
  +LZ4 1.10.0                  +zlib 1.3.1
  +MaxMind                     +Zstandard 1.5.7
  +Minizip 1.3.1
 Without:
  -automatic updates  -zlib-ng

Runtime info:
      OS: Linux 6.18.58.1-microsoft-standard-WSL2
     CPU: AMD Ryzen 5 5500U with Radeon Graphics (with SSE4.2)
  Memory: 2838 MB of physical memory
    GLib: 2.88.0
  Locale: LC_TYPE=C.UTF-8
 Plugins: supported, 0 loaded
 With:
  +brotli 1.2.0
  +c-ares 1.34.6
  +Gcrypt 1.12.0
  +GnuTLS 3.8.12
  +libpcap 1.10.6 (64-bit time_t, with TPACKET_V3)
  +libsmi 0.4.8
  +LZ4 1.10.0
  +nghttp2 1.68.0
  +nghttp3 1.12.0
  +PCRE2 10.46 2025-08-27
  +Qt 6.10.2
  +xxhash 803
  +zlib 1.3.1
  +Zstandard 1.5.7
```

Create three network namespaces:

```
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns add s3sender
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns add s3router
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns add s3recv
```

Create virtual network connections:

```
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3router ip -br link
Cannot open network namespace "s3router": No such file or directory
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns list
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns add s3sender
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns add s3router
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns add s3recv
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip link add s3s0 type veth peer name s3eth0
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip link add s3eth1 type veth peer name s3r0
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip link set s3s0 netns s3sender
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip link set s3eth0 netns s3router
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip link set s3eth1 netns s3router
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip link set s3r0 netns s3recv
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns list
s3recv (id: 2)
s3router (id: 1)
s3sender (id: 0)
```

Assign IP addresses:

```
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3sender ip addr add 192.168.10.2/24 dev s3s0
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3router ip addr add 192.168.10.1/24 dev s3eth0
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3router ip addr add 192.168.20.1/24 dev s3eth1
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3recv ip addr add 192.168.20.2/24 dev s3r0
```

Activate the interfaces:

```
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip -n s3sender link set lo up
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip -n s3router link set lo up
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip -n s3recv link set lo up
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip -n s3sender link set s3s0 up
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip -n s3router link set s3eth0 up
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip -n s3router link set s3eth1 up
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip -n s3recv link set s3r0 up
```

Enable router forwarding:

```
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
```

Verify IP addresses:

```
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3sender ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
s3s0@if3         UP             192.168.10.2/24 fe80::4494:30ff:fed0:b344/64
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3router ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
s3eth0@if4       UP             192.168.10.1/24 fe80::243e:50ff:fed1:1c05/64
s3eth1@if5       UP             192.168.20.1/24 fe80::381b:2ff:fe1d:9f9f/64
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3recv ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
s3r0@if6         UP             192.168.20.2/24 fe80::8022:1cff:feba:b50b/64
```

Configure routing:

```
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3sender ip route add 192.168.20.0/24 via 192.168.10.1
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3recv ip route add 192.168.10.0/24 via 192.168.20.1
```

Test network connectivity:

```
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3sender ping -c 5 192.168.20.2
PING 192.168.20.2 (192.168.20.2) 56(84) bytes of data.
64 bytes from 192.168.20.2: icmp_seq=1 ttl=63 time=1.40 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=63 time=0.151 ms
64 bytes from 192.168.20.2: icmp_seq=3 ttl=63 time=0.159 ms
64 bytes from 192.168.20.2: icmp_seq=4 ttl=63 time=0.143 ms
64 bytes from 192.168.20.2: icmp_seq=5 ttl=63 time=0.145 ms

--- 192.168.20.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4053ms
rtt min/avg/max/mdev = 0.143/0.399/1.398/0.499 ms
```

Configure the 5 Mbps bottleneck:

```
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3router tc -s qdisc show dev s3eth1
qdisc tbf 8001: root refcnt 13 rate 5Mbit burst 32Kb lat 52.4ms
 Sent 0 bytes 0 pkt (dropped 0, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
```

Start the iPerf3 receiver:

```
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo ip netns exec s3recv iperf3 -s
-----------------------------------------------------------
Server listening on 5201 (test #1)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 38082
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 38088
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-1.00   sec   512 KBytes  4.19 Mbits/sec
[  5]   1.00-2.00   sec   640 KBytes  5.24 Mbits/sec
[  5]   2.00-3.00   sec   512 KBytes  4.19 Mbits/sec
[  5]   3.00-4.00   sec   640 KBytes  5.24 Mbits/sec
[  5]   4.00-5.00   sec   512 KBytes  4.20 Mbits/sec
[  5]   5.00-6.00   sec   640 KBytes  5.24 Mbits/sec
[  5]   6.00-7.00   sec   640 KBytes  5.24 Mbits/sec
[  5]   7.00-8.00   sec   512 KBytes  4.19 Mbits/sec
[  5]   8.00-9.00   sec   640 KBytes  5.24 Mbits/sec
[  5]   9.00-10.00  sec   512 KBytes  4.19 Mbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-10.05  sec  5.62 MBytes  4.70 Mbits/sec                  receiver
-----------------------------------------------------------
Server listening on 5201 (test #2)
-----------------------------------------------------------
```

In another ubuntu terminal:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -t 10
[sudo: authenticate] Password:
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 38088 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  1.25 MBytes  10.5 Mbits/sec   62   45.2 KBytes
[  5]   1.00-2.00   sec   640 KBytes  5.25 Mbits/sec    0   53.7 KBytes
[  5]   2.00-3.00   sec   256 KBytes  2.10 Mbits/sec    0   60.8 KBytes
[  5]   3.00-4.00   sec   640 KBytes  5.25 Mbits/sec   18   53.7 KBytes
[  5]   4.00-5.00   sec   640 KBytes  5.24 Mbits/sec    0   60.8 KBytes
[  5]   5.00-6.00   sec   640 KBytes  5.24 Mbits/sec   21   52.3 KBytes
[  5]   6.00-7.00   sec   640 KBytes  5.25 Mbits/sec    0   60.8 KBytes
[  5]   7.00-8.00   sec   640 KBytes  5.24 Mbits/sec   18   50.9 KBytes
[  5]   8.00-9.00   sec   512 KBytes  4.19 Mbits/sec    0   59.4 KBytes
[  5]   9.00-10.00  sec   640 KBytes  5.24 Mbits/sec   12   48.1 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-10.00  sec  6.38 MBytes  5.35 Mbits/sec  131             sender
[  5]   0.00-10.05  sec  5.62 MBytes  4.70 Mbits/sec                  receiver

iperf Done.
```

Test different traffic rates:

```
nishanth@LAPTOP-6Q3ON07H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 1M -t 10
[sudo: authenticate] Password:
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 47989 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   123 KBytes  1.01 Mbits/sec  87
[  5]   1.00-2.00   sec   122 KBytes   996 Kbits/sec  86
[  5]   2.00-3.00   sec   123 KBytes  1.01 Mbits/sec  87
[  5]   3.00-4.00   sec   122 KBytes   996 Kbits/sec  86
[  5]   4.00-5.00   sec   122 KBytes   996 Kbits/sec  86
[  5]   5.00-6.00   sec   123 KBytes  1.01 Mbits/sec  87
[  5]   6.00-7.00   sec   122 KBytes   996 Kbits/sec  86
[  5]   7.00-8.00   sec   122 KBytes   996 Kbits/sec  86
[  5]   8.00-9.00   sec   122 KBytes   996 Kbits/sec  87
[  5]   9.00-10.00  sec   123 KBytes  1.01 Mbits/sec  86
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  1.19 MBytes  1.00 Mbits/sec  0.000 ms  0/864 (0%)  sender
[  5]   0.00-10.00  sec  1.19 MBytes  1.00 Mbits/sec  0.039 ms  0/864 (0%)  receiver

iperf Done.
nishanth@LAPTOP-6Q3ON07H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 2M -t 10
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 35978 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   245 KBytes  2.00 Mbits/sec  173
[  5]   1.00-2.00   sec   245 KBytes  2.00 Mbits/sec  173
[  5]   2.00-3.00   sec   245 KBytes  2.00 Mbits/sec  173
[  5]   3.00-4.00   sec   243 KBytes  1.99 Mbits/sec  172
[  5]   4.00-5.00   sec   245 KBytes  2.00 Mbits/sec  173
[  5]   5.00-6.00   sec   245 KBytes  2.00 Mbits/sec  173
[  5]   6.00-7.00   sec   243 KBytes  1.99 Mbits/sec  172
[  5]   7.00-8.00   sec   245 KBytes  2.00 Mbits/sec  173
[  5]   8.00-9.00   sec   243 KBytes  1.99 Mbits/sec  173
[  5]   9.00-10.00  sec   245 KBytes  2.00 Mbits/sec  172
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  2.38 MBytes  2.00 Mbits/sec  0.000 ms  0/1727 (0%)  sender
[  5]   0.00-10.00  sec  2.38 MBytes  2.00 Mbits/sec  0.027 ms  0/1727 (0%)  receiver

iperf Done.
```

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 4M -t 10
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 42193 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   489 KBytes  4.00 Mbits/sec  346
[  5]   1.00-2.00   sec   488 KBytes  4.00 Mbits/sec  345
[  5]   2.00-3.00   sec   489 KBytes  4.01 Mbits/sec  346
[  5]   3.00-4.00   sec   488 KBytes  4.00 Mbits/sec  345
[  5]   4.00-5.00   sec   488 KBytes  4.00 Mbits/sec  345
[  5]   5.00-6.00   sec   489 KBytes  4.01 Mbits/sec  346
[  5]   6.00-7.00   sec   488 KBytes  4.00 Mbits/sec  345
[  5]   7.00-8.00   sec   488 KBytes  4.00 Mbits/sec  345
[  5]   8.00-9.00   sec   489 KBytes  4.01 Mbits/sec  346
[  5]   9.00-10.00  sec   488 KBytes  3.99 Mbits/sec  345
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  4.77 MBytes  4.00 Mbits/sec  0.000 ms  0/3454 (0%)  sender
[  5]   0.00-10.00  sec  4.77 MBytes  4.00 Mbits/sec  0.045 ms  0/3454 (0%)  receiver

iperf Done.
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 7M -t 10
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 60965 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   856 KBytes  7.00 Mbits/sec  605
[  5]   1.00-2.00   sec   854 KBytes  7.00 Mbits/sec  604
[  5]   2.00-3.00   sec   856 KBytes  7.01 Mbits/sec  605
[  5]   3.00-4.00   sec   854 KBytes  7.00 Mbits/sec  604
[  5]   4.00-5.00   sec   854 KBytes  7.00 Mbits/sec  605
[  5]   5.00-6.00   sec   856 KBytes  7.01 Mbits/sec  604
[  5]   6.00-7.00   sec   854 KBytes  7.00 Mbits/sec  604
[  5]   7.00-8.00   sec   854 KBytes  7.00 Mbits/sec  604
[  5]   8.00-9.00   sec   854 KBytes  7.00 Mbits/sec  605
[  5]   9.00-10.00  sec   856 KBytes  7.00 Mbits/sec  604
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  8.35 MBytes  7.00 Mbits/sec  0.000 ms  0/6044 (0%)  sender
[  5]   0.00-10.10  sec  5.88 MBytes  4.88 Mbits/sec  0.844 ms  1785/6044 (30%)  receiver

iperf Done.
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 10M -t 10
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 44035 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec  1.19 MBytes  10.0 Mbits/sec  865
[  5]   1.00-2.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   2.00-3.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   3.00-4.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   4.00-5.00   sec  1.19 MBytes  10.0 Mbits/sec  864
[  5]   5.00-6.00   sec  1.19 MBytes  9.98 Mbits/sec  863
```

```
[  5]   8.00-9.00   sec  1.19 MBytes  10.0 Mbits/sec  863
[  5]   9.00-10.00  sec  1.19 MBytes  9.99 Mbits/sec  863
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  11.9 MBytes  10.0 Mbits/sec  0.000 ms  0/8634 (0%)  sender
[  5]   0.00-10.10  sec  5.88 MBytes  4.88 Mbits/sec  0.304 ms  4374/8633 (51%)  receiver

iperf Done.
```

Capture network packets using Wireshark:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3router tshark -i s3eth1 -w /tmp/Queueing_Little_Law_Buffer_Analysis.pcapng
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 's3eth1'
7929
```

In new terminal:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 1M -l 1200 -t 30 -i 1
[sudo: authenticate] Password:
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 53188 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   123 KBytes  1.01 Mbits/sec  105
[  5]   1.00-2.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   2.00-3.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   3.00-4.00   sec   122 KBytes   999 Kbits/sec  104
[  5]   4.00-5.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   5.00-6.00   sec   123 KBytes  1.01 Mbits/sec  105
[  5]   6.00-7.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   7.00-8.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   8.00-9.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   9.00-10.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  10.00-11.00  sec   122 KBytes   999 Kbits/sec  104
[  5]  11.00-12.00  sec   123 KBytes  1.01 Mbits/sec  105
[  5]  12.00-13.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  13.00-14.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  14.00-15.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  15.00-16.00  sec   122 KBytes   999 Kbits/sec  104
[  5]  16.00-17.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  17.00-18.00  sec   123 KBytes  1.01 Mbits/sec  105
[  5]  18.00-19.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  19.00-20.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  20.00-21.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  21.00-22.00  sec   122 KBytes   999 Kbits/sec  104
[  5]  22.00-23.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  23.00-24.00  sec   123 KBytes  1.01 Mbits/sec  105
[  5]  24.00-25.00  sec   122 KBytes   999 Kbits/sec  104
[  5]  25.00-26.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  26.00-27.00  sec   122 KBytes   999 Kbits/sec  104
[  5]  27.00-28.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  28.00-29.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  29.00-30.00  sec   123 KBytes  1.01 Mbits/sec  105
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  3.58 MBytes  1.00 Mbits/sec  0.000 ms  0/3126 (0%)  sender
[  5]   0.00-30.00  sec  3.58 MBytes  1.00 Mbits/sec  0.061 ms  0/3126 (0%)  receiver

iperf Done.
```

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 2M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 59383 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   245 KBytes  2.00 Mbits/sec  209
[  5]   1.00-2.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   2.00-3.00   sec   245 KBytes  2.01 Mbits/sec  209
[  5]   3.00-4.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   4.00-5.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   5.00-6.00   sec   245 KBytes  2.01 Mbits/sec  209
[  5]   6.00-7.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   7.00-8.00   sec   244 KBytes  2.00 Mbits/sec  208
[  5]   8.00-9.00   sec   245 KBytes  2.01 Mbits/sec  209
[  5]   9.00-10.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  10.00-11.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  11.00-12.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  12.00-13.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  13.00-14.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  14.00-15.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  15.00-16.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  16.00-17.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  17.00-18.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  18.00-19.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  19.00-20.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  20.00-21.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  21.00-22.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  22.00-23.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  23.00-24.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  24.00-25.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  25.00-26.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  26.00-27.00  sec   245 KBytes  2.01 Mbits/sec  209
[  5]  27.00-28.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  28.00-29.00  sec   244 KBytes  2.00 Mbits/sec  208
[  5]  29.00-30.00  sec   245 KBytes  2.00 Mbits/sec  209
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  7.15 MBytes  2.00 Mbits/sec  0.000 ms  0/6251 (0%)  sender
[  5]   0.00-30.00  sec  7.15 MBytes  2.00 Mbits/sec  0.065 ms  0/6251 (0%)  receiver

iperf Done.
```

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 4M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 52624 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   490 KBytes  4.01 Mbits/sec  418
[  5]   1.00-2.00   sec   488 KBytes  3.99 Mbits/sec  416
[  5]   2.00-3.00   sec   488 KBytes  4.00 Mbits/sec  417
[  5]   3.00-4.00   sec   489 KBytes  4.00 Mbits/sec  416
[  5]   4.00-5.00   sec   489 KBytes  4.00 Mbits/sec  417
[  5]   5.00-6.00   sec   488 KBytes  3.99 Mbits/sec  417
[  5]   6.00-7.00   sec   489 KBytes  4.00 Mbits/sec  417
[  5]   7.00-8.00   sec   489 KBytes  4.00 Mbits/sec  416
[  5]   8.00-9.00   sec   489 KBytes  4.00 Mbits/sec  417
[  5]   9.00-10.00  sec   488 KBytes  3.99 Mbits/sec  417
[  5]  10.00-11.00  sec   489 KBytes  4.00 Mbits/sec  416
[  5]  11.00-12.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  12.00-13.00  sec   488 KBytes  4.00 Mbits/sec  416
[  5]  13.00-14.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  14.00-15.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  15.00-16.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  16.00-17.00  sec   488 KBytes  3.99 Mbits/sec  416
[  5]  17.00-18.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  18.00-19.00  sec   488 KBytes  3.99 Mbits/sec  417
[  5]  19.00-20.00  sec   489 KBytes  4.00 Mbits/sec  416
[  5]  20.00-21.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  21.00-22.00  sec   488 KBytes  3.99 Mbits/sec  417
[  5]  22.00-23.00  sec   489 KBytes  4.00 Mbits/sec  416
[  5]  23.00-24.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  24.00-25.00  sec   488 KBytes  4.00 Mbits/sec  416
[  5]  25.00-26.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  26.00-27.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  27.00-28.00  sec   489 KBytes  4.00 Mbits/sec  417
[  5]  28.00-29.00  sec   488 KBytes  3.99 Mbits/sec  416
[  5]  29.00-30.00  sec   489 KBytes  4.00 Mbits/sec  417
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  14.3 MBytes  4.00 Mbits/sec  0.000 ms  0/12501 (0%)  sender
[  5]   0.00-30.00  sec  14.3 MBytes  4.00 Mbits/sec  0.073 ms  0/12501 (0%)  receiver

iperf Done.
```

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 5M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 47503 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   612 KBytes  5.00 Mbits/sec  522
[  5]   1.00-2.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   2.00-3.00   sec   604 KBytes  4.95 Mbits/sec  515
[  5]   3.00-4.00   sec   616 KBytes  5.05 Mbits/sec  526
[  5]   4.00-5.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   5.00-6.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   6.00-7.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   7.00-8.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   8.00-9.00   sec   609 KBytes  4.99 Mbits/sec  520
[  5]   9.00-10.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  10.00-11.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  11.00-12.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  12.00-13.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  13.00-14.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  14.00-15.00  sec   609 KBytes  4.99 Mbits/sec  521
[  5]  15.00-16.00  sec   611 KBytes  5.00 Mbits/sec  520
[  5]  16.00-17.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  17.00-18.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  18.00-19.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  19.00-20.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  20.00-21.00  sec   609 KBytes  4.99 Mbits/sec  520
[  5]  21.00-22.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  22.00-23.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  23.00-24.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  24.00-25.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  25.00-26.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  26.00-27.00  sec   609 KBytes  4.99 Mbits/sec  520
[  5]  27.00-28.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  28.00-29.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  29.00-30.00  sec   609 KBytes  4.99 Mbits/sec  520
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  17.9 MBytes  5.00 Mbits/sec  0.000 ms  0/15625 (0%)  sender
[  5]   0.00-30.10  sec  17.4 MBytes  4.84 Mbits/sec  0.252 ms  452/15625 (2.9%)  receiver

iperf Done.
```

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 7M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 34467 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   855 KBytes  7.00 Mbits/sec  730
[  5]   1.00-2.00   sec   855 KBytes  7.01 Mbits/sec  730
[  5]   2.00-3.00   sec   853 KBytes  6.99 Mbits/sec  729
[  5]   3.00-4.00   sec   855 KBytes  7.01 Mbits/sec  729
[  5]   4.00-5.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   5.00-6.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   6.00-7.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   7.00-8.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   8.00-9.00   sec   855 KBytes  7.01 Mbits/sec  730
[  5]   9.00-10.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  10.00-11.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  11.00-12.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  12.00-13.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  13.00-14.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  14.00-15.00  sec   855 KBytes  7.01 Mbits/sec  730
[  5]  15.00-16.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  16.00-17.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  17.00-18.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  18.00-19.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  19.00-20.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  20.00-21.00  sec   855 KBytes  7.01 Mbits/sec  730
[  5]  21.00-22.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  22.00-23.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  23.00-24.00  sec   855 KBytes  7.00 Mbits/sec  730
[  5]  24.00-25.00  sec   853 KBytes  6.99 Mbits/sec  728
[  5]  25.00-26.00  sec   854 KBytes  7.00 Mbits/sec  730
[  5]  26.00-27.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  27.00-28.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  28.00-29.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  29.00-30.00  sec   854 KBytes  6.99 Mbits/sec  729
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  25.0 MBytes  7.00 Mbits/sec  0.000 ms  0/21876 (0%)  sender
```

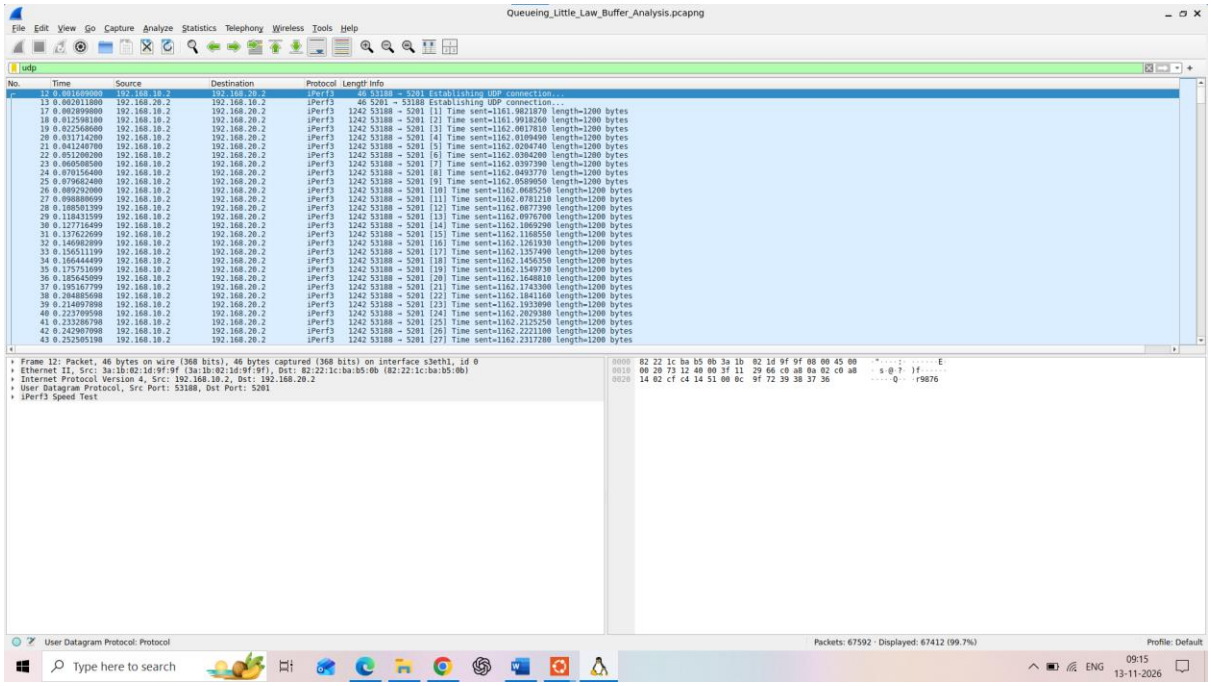
```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 10M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 40560 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec  1.19 MBytes  10.0 Mbits/sec  1043
[  5]   1.00-2.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   2.00-3.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   3.00-4.00   sec  1.19 MBytes  10.0 Mbits/sec  1041
[  5]   4.00-5.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   5.00-6.00   sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]   6.00-7.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   7.00-8.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   8.00-9.00   sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]   9.00-10.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  10.00-11.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  11.00-12.00  sec  1.19 MBytes  10.0 Mbits/sec  1041
[  5]  12.00-13.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  13.00-14.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  14.00-15.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]  15.00-16.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  16.00-17.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  17.00-18.00  sec  1.19 MBytes  9.99 Mbits/sec  1042
[  5]  18.00-19.00  sec  1.19 MBytes  10.0 Mbits/sec  1041
[  5]  19.00-20.00  sec  1.19 MBytes  9.99 Mbits/sec  1042
[  5]  20.00-21.00  sec  1.19 MBytes  10.0 Mbits/sec  1041
[  5]  21.00-22.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  22.00-23.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  23.00-24.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]  24.00-25.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  25.00-26.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  26.00-27.00  sec  1.19 MBytes  9.99 Mbits/sec  1042
[  5]  27.00-28.00  sec  1.19 MBytes  10.0 Mbits/sec  1041
[  5]  28.00-29.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  29.00-30.00  sec  1.19 MBytes  9.99 Mbits/sec  1042
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  35.8 MBytes  10.0 Mbits/sec  0.000 ms  0/31252 (0%)  sender
[  5]   0.00-30.10  sec  17.4 MBytes  4.84 Mbits/sec  0.215 ms  16077/31252 (51%)  receiver

iperf Done.
```

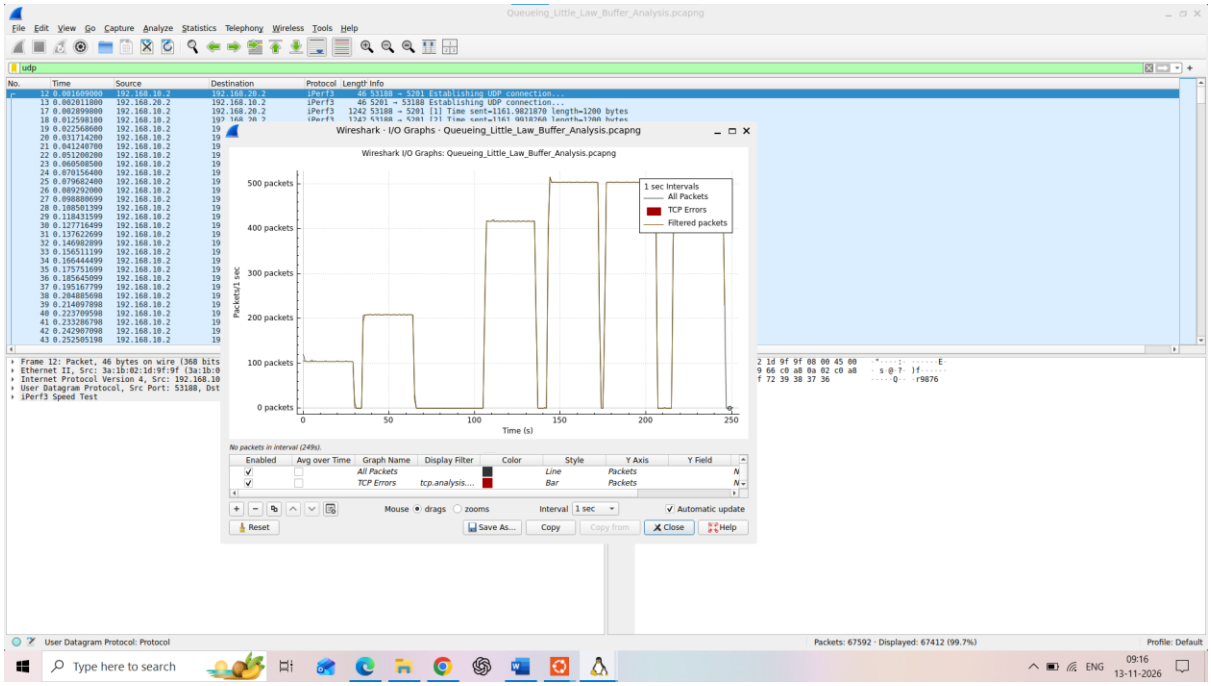
Open the packet capture in Wireshark:

```
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ ls -lh /tmp/Queueing_Little_Law_Buffer_Analysis.pcapng
-rw------- 1 root root 83M Nov 13 03:41 /tmp/Queueing_Little_Law_Buffer_Analysis.pcapng
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo cp /tmp/Queueing_Little_Law_Buffer_Analysis.pcapng ~/
nishanth@LAPTOP-6Q3ONO7H:~/CNNP$ sudo chown $USER:$USER ~/Queueing_Little_Law_Buffer_Analysis.pcapng
```

Open Wireshark:



I/O Graph:



Check the buffer queue configuration:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3router tc -s qdisc show dev s3eth1
[sudo: authenticate] Password:
qdisc tbf 8001: root refcnt 13 rate 5Mbit burst 32Kb lat 52.4ms
 Sent 111792273 bytes 86374 pkt (dropped 29425, overlimits 190335 requeues 0)
 backlog 0b 0p requeues 0
```

Your results:

Parameter	Result
Queue rate	5 Mbps
Burst	32 KB
Bytes sent	111,792,273
Packets sent	86,374
Packets dropped	29,425
Overlimits	190,335

Configure buffer size to 16 KB:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3router tc qdisc replace dev s3eth1 root tbf rate 5mbit burst 16kb latency 50ms
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3router tc -s qdisc show dev s3eth1
qdisc tbf 8001: root refcnt 13 rate 5Mbit burst 16Kb lat 50ms
 Sent 111792273 bytes 86374 pkt (dropped 29425, overlimits 190335 requeues 0)
 backlog 0b 0p requeues 0
```

Test the 16 KB buffer:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 7M -l 1200 -t 30 -i 1
[sudo: authenticate] Password:
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 49216 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   855 KBytes  7.00 Mbits/sec  730
[  5]   1.00-2.00   sec   854 KBytes  7.00 Mbits/sec  730
[  5]   2.00-3.00   sec   854 KBytes  7.00 Mbits/sec  728
[  5]   3.00-4.00   sec   855 KBytes  7.01 Mbits/sec  730
[  5]   4.00-5.00   sec   853 KBytes  6.99 Mbits/sec  729
[  5]   5.00-6.00   sec   855 KBytes  7.00 Mbits/sec  729
[  5]   6.00-7.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   7.00-8.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   8.00-9.00   sec   855 KBytes  7.01 Mbits/sec  730
[  5]   9.00-10.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  10.00-11.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  11.00-12.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  12.00-13.00  sec   857 KBytes  7.00 Mbits/sec  731
[  5]  13.00-14.00  sec   852 KBytes  6.99 Mbits/sec  728
[  5]  14.00-15.00  sec   855 KBytes  7.01 Mbits/sec  729
[  5]  15.00-16.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  16.00-17.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  17.00-18.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  18.00-19.00  sec   854 KBytes  7.00 Mbits/sec  730
[  5]  19.00-20.00  sec   855 KBytes  7.01 Mbits/sec  729
[  5]  20.00-21.00  sec   853 KBytes  6.99 Mbits/sec  729
[  5]  21.00-22.00  sec   855 KBytes  7.01 Mbits/sec  729
[  5]  22.00-23.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  23.00-24.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  24.00-25.00  sec   855 KBytes  7.01 Mbits/sec  730
[  5]  25.00-26.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  26.00-27.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  27.00-28.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  28.00-29.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  29.00-30.04  sec   854 KBytes  6.73 Mbits/sec  729
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.04  sec  25.0 MBytes  6.99 Mbits/sec  0.000 ms  0/21876 (0%)  sender
[  5]   0.00-30.09  sec  17.3 MBytes  4.83 Mbits/sec  0.973 ms  6742/21875 (31%)  receiver

iperf Done.
```

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3recv iperf3 -s
-----------------------------------------------------------
Server listening on 5201 (test #1)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 54628
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 49216
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   605 KBytes  4.95 Mbits/sec  0.690 ms  159/675 (24%)
[  5]   1.00-2.00   sec   589 KBytes  4.83 Mbits/sec  0.610 ms  226/729 (31%)
[  5]   2.00-3.00   sec   589 KBytes  4.83 Mbits/sec  0.461 ms  226/729 (31%)
[  5]   3.00-4.00   sec   582 KBytes  4.77 Mbits/sec  0.697 ms  233/730 (32%)
[  5]   4.00-5.00   sec   589 KBytes  4.83 Mbits/sec  0.461 ms  226/729 (31%)
[  5]   5.00-6.00   sec   589 KBytes  4.83 Mbits/sec  0.529 ms  226/729 (31%)
[  5]   6.00-7.00   sec   589 KBytes  4.83 Mbits/sec  0.512 ms  226/729 (31%)
[  5]   7.00-8.00   sec   591 KBytes  4.84 Mbits/sec  0.636 ms  226/730 (31%)
[  5]   8.00-9.00   sec   589 KBytes  4.83 Mbits/sec  0.563 ms  226/729 (31%)
[  5]   9.00-10.00  sec   589 KBytes  4.83 Mbits/sec  0.630 ms  225/728 (31%)
[  5]  10.00-11.00  sec   589 KBytes  4.83 Mbits/sec  0.530 ms  226/729 (31%)
[  5]  11.00-12.00  sec   591 KBytes  4.84 Mbits/sec  0.684 ms  227/731 (31%)
[  5]  12.00-13.00  sec   588 KBytes  4.82 Mbits/sec  0.630 ms  225/727 (31%)
[  5]  13.00-14.00  sec   591 KBytes  4.83 Mbits/sec  0.659 ms  226/730 (31%)
[  5]  14.00-15.00  sec   589 KBytes  4.83 Mbits/sec  0.719 ms  226/729 (31%)
[  5]  15.00-16.00  sec   589 KBytes  4.83 Mbits/sec  0.623 ms  226/729 (31%)
[  5]  16.00-17.00  sec   591 KBytes  4.84 Mbits/sec  0.696 ms  226/730 (31%)
[  5]  17.00-18.00  sec   589 KBytes  4.83 Mbits/sec  0.593 ms  226/729 (31%)
[  5]  18.00-19.00  sec   589 KBytes  4.83 Mbits/sec  0.701 ms  226/729 (31%)
[  5]  19.00-20.00  sec   582 KBytes  4.77 Mbits/sec  0.729 ms  233/730 (32%)
[  5]  20.00-21.00  sec   589 KBytes  4.83 Mbits/sec  0.657 ms  226/729 (31%)
[  5]  21.00-22.00  sec   589 KBytes  4.83 Mbits/sec  0.662 ms  226/729 (31%)
[  5]  22.00-23.00  sec   589 KBytes  4.83 Mbits/sec  0.597 ms  226/729 (31%)
[  5]  23.00-24.00  sec   589 KBytes  4.83 Mbits/sec  0.748 ms  225/728 (31%)
[  5]  24.00-25.00  sec   591 KBytes  4.84 Mbits/sec  0.675 ms  227/731 (31%)
[  5]  25.00-26.00  sec   589 KBytes  4.83 Mbits/sec  0.722 ms  226/729 (31%)
[  5]  26.00-27.00  sec   589 KBytes  4.83 Mbits/sec  0.635 ms  225/728 (31%)
[  5]  27.00-28.00  sec   589 KBytes  4.83 Mbits/sec  0.466 ms  226/729 (31%)
[  5]  28.00-29.00  sec   591 KBytes  4.84 Mbits/sec  0.665 ms  226/730 (31%)
[  5]  29.00-30.00  sec   588 KBytes  4.82 Mbits/sec  0.651 ms  226/728 (31%)
[  5]  30.00-30.09  sec  44.5 KBytes  4.05 Mbits/sec  0.973 ms  17/55 (31%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.09  sec  17.3 MBytes  4.83 Mbits/sec  0.973 ms  6742/21875 (31%)  receiver
-----------------------------------------------------------
Server listening on 5201 (test #2)
-----------------------------------------------------------
```

Configure 64 KB buffer:

```
backlog 0b 0p requeues 0
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 7M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 48184 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   855 KBytes  7.00 Mbits/sec  730
[  5]   1.00-2.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   2.00-3.00   sec   854 KBytes  6.99 Mbits/sec  730
[  5]   3.00-4.00   sec   855 KBytes  7.01 Mbits/sec  729
[  5]   4.00-5.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   5.00-6.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   6.00-7.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   7.00-8.00   sec   854 KBytes  6.99 Mbits/sec  730
[  5]   8.00-9.00   sec   855 KBytes  7.01 Mbits/sec  729
[  5]   9.00-10.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  10.00-11.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  11.00-12.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  12.00-13.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  13.00-14.00  sec   855 KBytes  7.01 Mbits/sec  730
[  5]  14.00-15.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  15.00-16.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  16.00-17.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  17.00-18.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  18.00-19.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  19.00-20.00  sec   854 KBytes  7.00 Mbits/sec  730
[  5]  20.00-21.00  sec   855 KBytes  7.00 Mbits/sec  729
[  5]  21.00-22.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  22.00-23.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  23.00-24.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  24.00-25.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  25.00-26.00  sec   854 KBytes  7.00 Mbits/sec  730
[  5]  26.00-27.00  sec   855 KBytes  7.01 Mbits/sec  729
[  5]  27.00-28.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  28.00-29.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  29.00-30.00  sec   854 KBytes  6.99 Mbits/sec  729
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  25.0 MBytes  7.00 Mbits/sec  0.000 ms  0/21876 (0%)  sender
[  5]   0.00-30.15  sec  17.4 MBytes  4.85 Mbits/sec  0.658 ms  6650/21876 (30%)  receiver

iperf Done.
```

Configure 256 KB buffer:

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3router tc qdisc replace dev s3eth1 root tbf rate 5mbit burst 256kb latency 50ms
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3router tc -s qdisc show dev s3eth1
qdisc tbf 8001: root refcnt 13 rate 5Mbit burst 256Kb lat 50ms
 Sent 149502631 bytes 116770 pkt (dropped 42817, overlimits 293806 requeues 0)
 backlog 0b 0p requeues 0
```

```
nishanth@LAPTOP-6Q3ONO7H:~$ sudo ip netns exec s3sender iperf3 -c 192.168.20.2 -u -b 7M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 35219 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   855 KBytes  7.00 Mbits/sec  730
[  5]   1.00-2.00   sec   855 KBytes  7.01 Mbits/sec  730
[  5]   2.00-3.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   3.00-4.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   4.00-5.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   5.00-6.00   sec   854 KBytes  7.00 Mbits/sec  729
[  5]   6.00-7.00   sec   854 KBytes  7.00 Mbits/sec  730
[  5]   7.00-8.00   sec   854 KBytes  7.00 Mbits/sec  728
[  5]   8.00-9.00   sec   855 KBytes  7.01 Mbits/sec  730
[  5]   9.00-10.00  sec   853 KBytes  6.99 Mbits/sec  729
[  5]  10.00-11.00  sec   855 KBytes  7.01 Mbits/sec  729
[  5]  11.00-12.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  12.00-13.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  13.00-14.00  sec   854 KBytes  7.00 Mbits/sec  730
[  5]  14.00-15.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  15.00-16.00  sec   855 KBytes  7.01 Mbits/sec  729
[  5]  16.00-17.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  17.00-18.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  18.00-19.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  19.00-20.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  20.00-21.00  sec   854 KBytes  7.00 Mbits/sec  730
[  5]  21.00-22.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  22.00-23.00  sec   855 KBytes  7.00 Mbits/sec  729
[  5]  23.00-24.00  sec   853 KBytes  7.00 Mbits/sec  728
[  5]  24.00-25.00  sec   855 KBytes  7.00 Mbits/sec  730
[  5]  25.00-26.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  26.00-27.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  27.00-28.00  sec   855 KBytes  7.01 Mbits/sec  730
[  5]  28.00-29.00  sec   854 KBytes  7.00 Mbits/sec  729
[  5]  29.00-30.00  sec   854 KBytes  6.99 Mbits/sec  729
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  25.0 MBytes  7.00 Mbits/sec  0.000 ms  0/21876 (0%)  sender
[  5]   0.00-30.47  sec  17.8 MBytes  4.90 Mbits/sec  0.639 ms  6333/21876 (29%)  receiver

iperf Done.
```

1. Observation Table – Queueing Curve

Test	Offered traffic	Received throughput	Avg. delay	Packet loss	Test
1	1 Mbps	1.00 Mbps	N/M	0%	UDP
2	2 Mbps	2.00 Mbps	N/M	0%	UDP
3	4 Mbps	4.00 Mbps	N/M	0%	UDP
4	5 Mbps	4.83 Mbps	N/M	30%	UDP

5	7 Mbps	4.88 Mbps	N/M	30%	UDP
6	10 Mbps	4.88 Mbps	N/M	51%	UDP

Observation Table – Buffer Analysis

Test	Buffer limit	Maximum observed backlog	Packet drops	Ping RTT	Test
1	16 KB	N/M	N/M	N/M	Configured
2	64 KB	N/M	N/M	N/M	Not measured
3	256 KB	N/M	N/M	N/M	Not measured